KATO KSP30 Centrifugal Pump coupled to a KATO KD80-N Diesel Engine

Test Report No.	:	2014 - 25
Date of Test	:	March 11, 2014
Test Valid Until	:	March 11, 2019
Brand	:	KATO
Model	:	KSP30
Serial No.	:	No Data
Type and Size	:	Single-end suction, 75 x 75 mm, Non-Self Priming Centrifugal pump
Manufacturer	:	KATO METAL CORP. Valenzuela City
Test Requested by	:	TRAMAT MERCANTILE, INC. 747-749 Sabino Padilla St., Sta. Cruz, Manila
Machine Classification	:	Commercial

SUMMARY

The KATO KSP30 centrifugal pump coupled to a KATO KD80-N diesel engine was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump both used double groove B-section pulleys with actual diameters of 102 mm (4.0 inches) and 88.9 mm (3.5 inches), respectively, and two pieces of V-belts. The engine and pump were mounted on a steel base.

The Maximum Total Head (H) was 31.0 m at zero discharge condition. The shaft speed of the pump and engine at this point were 2047 and 2400 rpm, respectively with Engine Fuel Consumption Rate of 2.53 L/h. The Maximum Discharge Capacity of 14.6 L/s occurred at 7.54 m Total Head with Fuel Consumption Rate of 1.45 L/h. The shaft speed of the pump and engine at this point were 1920 and 1717 rpm respectively.

The Maximum System Efficiency of 9.13% occurred at 9.84 L/s Discharge Capacity.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.56 m. The Net Positive Suction Head Available at this point was 2.32 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pump performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pump was mounted on the laboratory test set-up shown in **Figure 1.**

The test was conducted on March 11, 2014 at the AMTEC Test Laboratory, UPLB, College, Laguna.

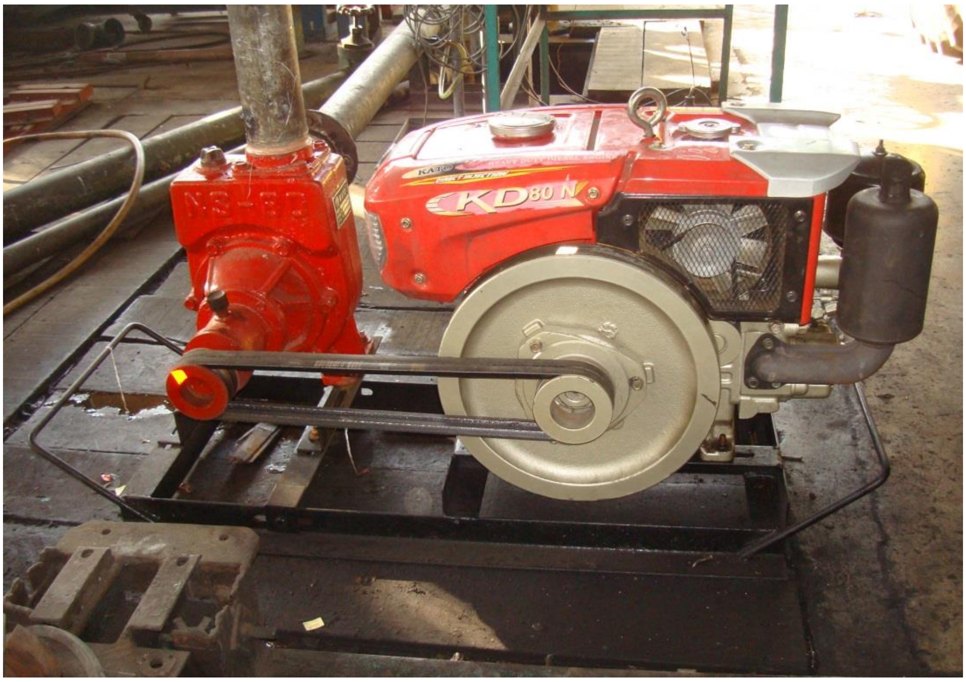

Figure 1. KATO KSP30 Centrifugal pump coupled to KATO KD80-N Diesel Engine on the test set-up

METHOD OF TEST

The pump was tested following the "Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test".

DESCRIPTION OF THE PUMP

The KATO KSP30 Centrifugal pump coupled to KATO KD80-N Diesel Engine shown in **Figure 2**, was a single-end suction, 75 mm x 75 mm, centrifugal pump coupled to 6.25 kW (8.38 Hp) KATO KD80-N diesel engine. The pump and engine were mounted on a steel base. The power from the engine was transmitted to the pump shaft through a double groove B-section pulley and V-belts. The actual diameters of the engine and pump pulleys were 102 mm (4.0 inches) and 88.9 mm (3.5 inches), respectively. The suction inlet and discharge outlet were threaded and designed for a 75 mm diameter pipe. The impeller was a semi-open type made of brass material connected to the pump shaft through a thread. It was housed in a cast iron volute casing. The priming inlet was located on top of the pump beside the discharge outlet. The discharge outlet was located on top of the pump facing upward. The stuffing box used a packing seal. The pump shaft rotated on ball bearing kept in place by a bearing housing integral to the casing.

The prime mover, a 6.25 kW KATO KD80-N diesel engine was a four-stroke, water-cooled, single horizontal cylinder diesel engine. Fuel feed system was of gravity type with the fuel tank located on top of the engine. It utilized an oil-bath type air cleaner. Lubrication system was force-feed lubricating type. Cooling was done by water circulating from cylinder jacket to the radiator using thermo-siphon principle. A fan driven by the pulley on the flywheel was used to cool the warm water in the radiator. It was started using manual cranking lever aided by a manually actuated decompression lever. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.

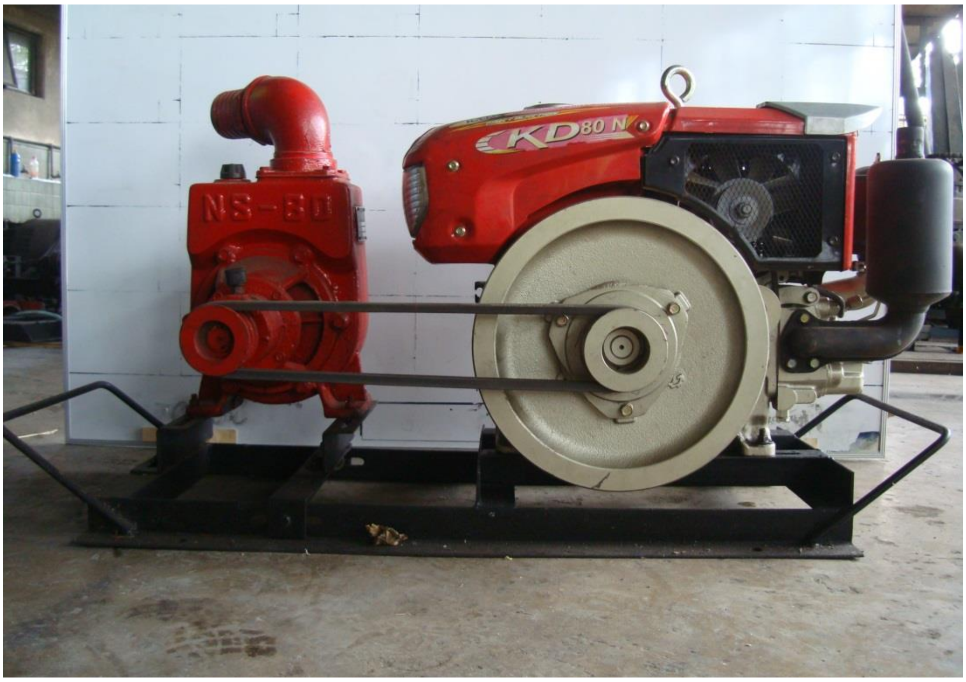

Figure 2. KATO KSP30 Centrifugal Pump coupled to a KATO KD80-N Diesel Engine

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the KATO KSP30 Centrifugal Pump coupled to KATO KD80-N Diesel Engine

1.	Brand		:	KATO
2.	Model		:	KSP30
3.	Serial No.		:	No Data
4.	Make		:	KATO METAL CORP. Valenzuela City
5.	Type		:	Single-end suction, Centrifugal pump
6.	Size (mm)		:	75 x 75
7.	Overall Dimension and Weight (Including mounting frame and engine)			
	7.1	Length (mm)	:	1215
	7.2	Width (mm)	:	460
	7.3	Height (mm)	:	560
	7.4	Weight (kg)	:	144
8.	Impeller			
	8.1	Type	:	Semi-open
	8.2	Diameter (mm)	:	185
	8.3	Width (mm)	:	31
	8.4	No. of vanes	:	5
	8.5	Material	:	Brass
9.	Suction Inlet			
	9.1	Type	:	Threaded flange
	9.2	Diameter (mm)	:	75
	9.3	Material	:	Cast Iron

Table 1. (Cont'n)

10.	Discharge Outlet		
	10.1 Type	:	Threaded flange
	10.2 Diameter (mm)	:	75
	10.3 Material	:	Cast Iron
11.	Casing		
	11.1 Type	:	Volute
	11.2 Material	:	Cast Iron
12	Stuffing box	:	Packing Seal
13.	Rotation	:	Clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	2400
15.	Rated Maximum Discharge Capacity (L/s)	:	No Data
16.	Rated Maximum Total Head (m)	:	No Data
17.	Prime mover (based on Plate Rating)		
	17.1 Brand	:	KATO
	17.2 Model	:	KD80-N
	17.3 Serial No.	:	No Data
	17.4 Make	:	GOLDEN FLYING FISH DIESEL ENGINE CO., LTD.
	17.5 Type	:	Single cylinder, 4-stroke, water-cooled, horizontal cylinder diesel engine
	17.6 Rated Output Power (kW)		
	Maximum	:	6.25 @ 2400 rpm
	Continuous	:	No Data

Table 1. (Cont'n)

17.7	Bore x Stroke (mm)	:	No Data
17.8	Displacement Volume (cc)	:	487
17.9	Cooling System	:	Water-cooled
17.10	Lubricating System	:	Force-feed
17.11	Starting System	:	Manual Cranking
17.12	Air Cleaner	:	Oil Bath type
17.13	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pump Tested : KATO KSP30 coupled to KATO KD80-N diesel engine

Date of Test : March 11, 2014

Test Condition

Ambient Temperature (^{o}C) : 26.4
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (^{o}C) : 29.4

Table 2. Results of Performance Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	FUEL CONSUMP-TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY (%)
2400	2047	2.53	31.0	0	0	26.3	0
2412	2467	2.06	26.9	3.92	1.03	21.4	4.81
2419	2294	2.0	22.3	5.80	1.27	20.8	6.09
2385	2270	2.16	19.6	8.03	1.54	22.5	6.88
2093	2189	1.84	18.0	9.10	1.60	19.1	8.40
1863	2101	1.56	15.3	9.84	1.48	16.2	9.13
1804	2033	1.51	12.7	11.4	1.42	15.7	9.08
1786	2008	1.49	11.3	11.8	1.31	15.5	8.46
1738	1961	1.68	9.50	13.6	1.27	17.5	7.26
1717	1920	1.45	7.54	14.6	1.08	15.1	7.18

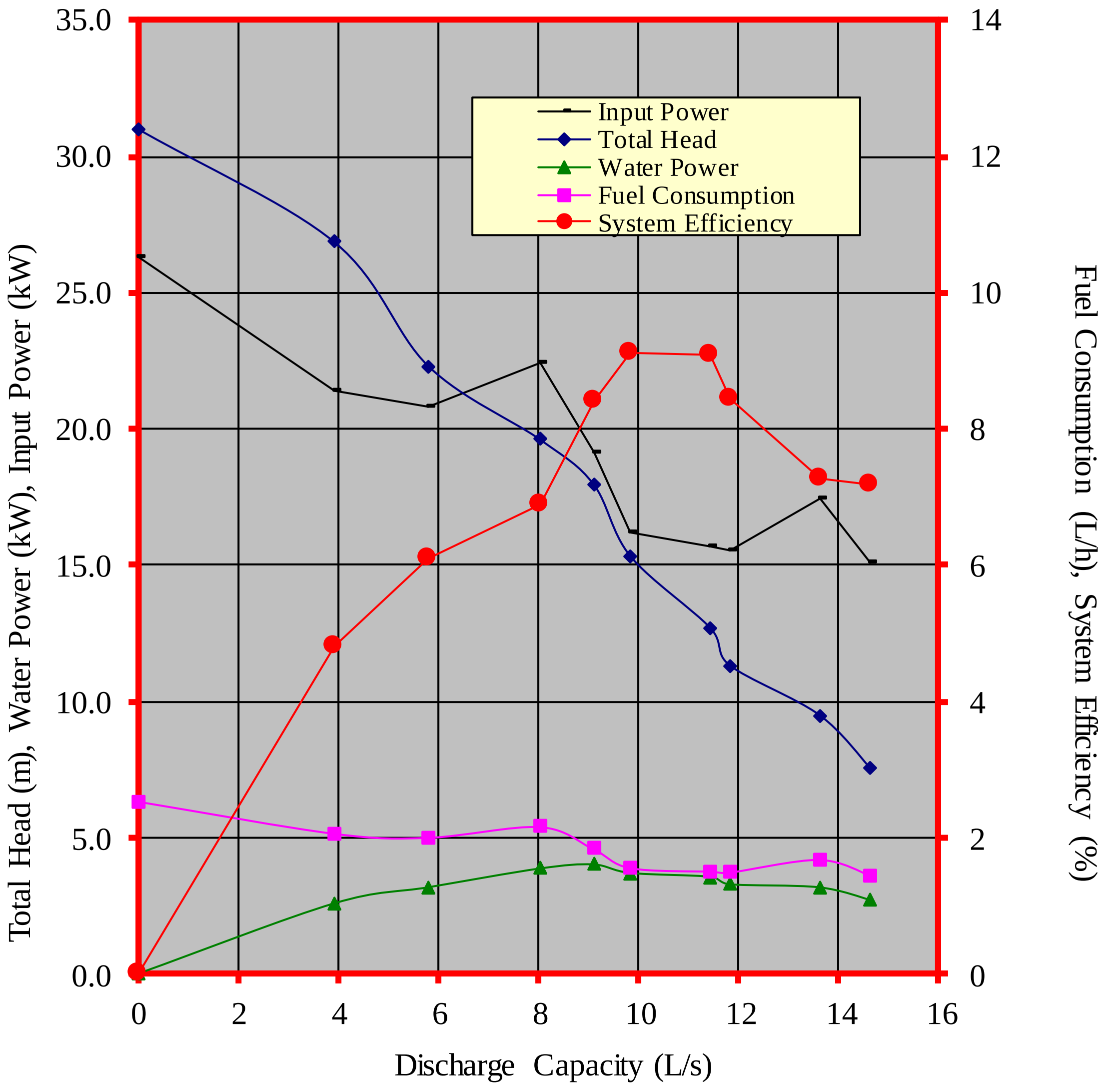

Figure 3. Performance characteristic curves of the KATO KSP30 centrifugal pump coupled to KATO KD80-N diesel engine

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pump Tested : KATO KSP30 coupled to
KATO KD80-N diesel engine

Date of Test : March 11, 2014

Test Condition

Ambient Temperature (^{o}C) : 30.5
Atmospheric Pressure (mb) : 1013
Pumping Water Temperature (^{o}C) : 29.0

Table 3. Results of Cavitation Test

ENGINE SHAFT SPEED (rpm)	PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
1730	1946	1.19	8.68	45.0	16.3
1703	1915	1.55	8.32	39.2	15.1
1730	1939	2.98	6.89	33.8	14.1
1740	1952	4.54	5.33	31.0	10.9
1815	2033	5.87	4.0	30.5	11.5
2053	2320	6.77	3.10	29.0	6.49
2426	2730	7.56	2.32	26.7	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

Where: $$H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted to meter. Computed by determining the velocity of liquid on the pipe by formula, V = Q/A where V, velocity of liquid inside the pipe in meter per second, Q, Discharge rate of the pump in liters per second, and A, Cross-sectional area of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q, in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power, WP, is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power, IP, is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters
P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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N.B.

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